

Assignment 2

Watch the following videos:

1. The speaker Steve Brunton has pioneered the use of machine learning to fluid mechanics. He is a co-author on 'Data-Driven Fluid Mechanics: Combining First Principles and Machine Learning'
<https://www.youtube.com/watch?v=Jt5R-Tm8cV8>
<https://www.youtube.com/watch?v=lKXW7DRyyro>

Optional: You can also look at his ML(and Differential Equation) algorithm SINDY
https://en.wikipedia.org/wiki/Sparse_identification_of_non-linear_dynamics#cite_ref-1
<https://www.youtube.com/watch?v=NxAn0ogIMVw>

2. 3Blue1Brown:
<https://www.youtube.com/watch?v=rB83DpBJQsE>
3. Vector calculus:
<https://www.youtube.com/watch?v=AMQ6VeXV278>

After watching the above try to briefly answer the following questions.

Question 1:

Write down the mathematical operation for the nabla (del) operator? Is it a scalar operator or a vector?

Question 2:

What is a field? Differentiate between a scalar and a vector field with real world examples.

Question 3:

Take the fields

$$f(x,y,z) = 2x + 3xy + 2z^2$$
$$g(x,y,z) = 2x \mathbf{i} + 3xy \mathbf{j} + 2z^2 \mathbf{k}$$

Can you find (if possible) at point (1,2,3):

- The gradient of both fields?
- The divergence of both fields?
- The curl of both fields?

Question 4

Solve question 3 using python.

Congratulations you now know (differential) vector calculus. To understand integral vector calculus we will have a similar assignment on:

- Divergence theorem
- Stokes' Theorem (which is a general case of Green's Theorem)